DATA 255 Deep Learning Technologies

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Interdisciplinary Science 784

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SJSU SAN JOSÉ STATE UNIVERSITY

Instructor

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Office hours

- By Walk-in/Appointment

- ISB866 between 5pm and 5:50pm on Monday/Thursday
- or by Appointment
- Or Zoom

Instructional Student Assistant (ISA)

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Courses for Advanced Data Science

Course	Title
DATA 200	Computational Programming for Data Analytics
DATA 201	Database Technologies for Data Analytics
DATA 202	Mathematics for Applied Data Science
DATA 220	Mathematical Methods for Data Analytics
DATA 226	Data Warehouse and Pipeline
DATA 228	Big Data Technologies and Applications
DATA 230	Business Intelligence and Data Visualization
DATA 236	Distributed Systems for Data Engineering
DATA 245	Machine Learning Technologies
DATA 255	Deep Learning Technologies
DATA 265	Large Language Model Applications
DATA 266	Generative AI Applications

Course Description and information

Study deep learning theory, techniques and technologies implement neural networks using deep learning frameworks to solve problems in various application domains

- Fundamentals of Neural network
- Deep neural network (DNN)
- Convolutional neural networks (CNN)
- Autoencoder and VAE
- Generative adversarial networks (GAN)
- Diffusion models

- Prerequisite(s):
  - DATA 245 (Machine Learning Technologies): with a “C” or better grade

- Corequisite(s):
  - DATA 266 (Generative Model Applications)

Course Protocol

- Punctuality:
  - Arrive on time.
- Attendance Responsibility:
  - If you miss a class, it is your responsibility to catch up on any announcements, changes to deadlines, or materials covered.
- Device Etiquette:
  - Turn off or mute your cell phone during class.
- Classroom Conduct:
  - Refrain from engaging in activities that are irrelevant or distracting to others.
- Active Participation:
  - Stay engaged and contribute to discussions and activities to enhance the learning experience for everyone.

Course Protocol

- Students Are Not Allowed to Record

  Students are prohibited from recording class activities (including class lectures, office hours, advising sessions, etc.), distributing class recordings, or posting class recordings.

  Materials created by the instructor for the course (syllabi, lectures and lecture notes, presentations, etc.) are copyrighted by the instructor.

  This university policy (S12-7) is in place to protect the privacy of students in the course, as well as to maintain academic integrity by reducing the instances of cheating. Students who record, distribute, or post these materials will be referred to the Student Conduct and Ethical Development office. Unauthorized recording may violate university and state law.

  It is the responsibility of students that require special accommodations or assistive technology due to a disability to notify the Accessible Education Center(AEC) at http://www.sjsu.edu/aec. AEC will in turn contact the instructor as needed.

Textbook

This class does not have a required textbook. Lecture slides and selected readings from online materials will cover most of the topics.

Reference textbooks (not required)

- Math and Architectures of Deep Learning, Krishenedu Chauhury
- Hands-On Machine Learning with Scikit-Learn & TensorFlow, Aurelien Geron
- Deep Learning with Python, Francois Chollet
- Neural Networks and Deep Learning, Charu C. Aggarwal
- Introduction to Deep Learning, S. Skansi
- Deep Learning for Vision Systems, Mohamed Elgendy
- Generative Deep Learning(2^{nd} Ed), David Foster
- GANs in action, Jakub Langr & Vladimir Bok
- Hands-On Generative AI with Transformers and Diffusion Models, Omar Sanseviero et al.

Course Prerequisites

- Proficiency in Python
  - The majority of class assignments will be in Python. You should be familiar with Jupiter notebook and Pytorch.
- College Calculus, Linear Algebra
  - You should be comfortable understanding matrix/vector notation and operations
- Basic Probability and Statistics
  - You should know the basics of probabilities, Gaussian distributions, mean, standard deviation, etc.
- Machine Learning
  - You should be comfortable with the basics of machine learning

Academic integrity: Following is prohibited

- Copying from another's work or allowing someone to copy from your own work
- Representing someone else's work as your own
- Plagiarism

Copying and other forms of cheating will not be tolerated and will result in a zero score for the quiz, assignment, exam (minimal penalty) or a failing grade for the course, possibly combined with other disciplinary actions from the university.

From University Policy F15-7

1.2 PLAGIARISM

San José State University defines plagiarism as the act of representing the work of another as one's own without giving appropriate credit, regardless of how that work was obtained, and submitting it to fulfill academic requirements.

Plagiarism includes:

1.2.1 Knowingly or unknowingly incorporating the ideas, words, sentences, paragraphs, or parts of sentences or paragraphs, or the specific substance of another's work without giving appropriate credit, and representing the product as one's own work;

1.2.2 Representing another's artistic or scholarly works, such as computer programs, instrument printouts, inventions, musical compositions, photographs, paintings, drawings, sculptures, novels, short stories, poems, screen plays, or television scripts, as one's own.

AI tools

- Prompting LLMs such as ChatGPT is permitted for low-level programming questions or high-level conceptual questions about language models but using it directly to solve the problem is prohibited.
- AI-generated solutions will not be accepted and will be considered cheating.
- If AI use is necessary for any assignment, it will be announced when and how it can be used.

Determination of Grades

In-class quizzes	10 %
Assignments	10 %
Labs	20 %
Group Project	20 %
Mid-term exam	15 %
Final exam	25 %

Grading criteria of quizzes, Assignments, Labs, Mid-term exam, Final exam would be different.

Grading breakdown

The final grades will be calculated based on the following:

(A+) >= 96%,

(A) >= 93% and <96%

(A-) >= 90% and <93%

(B+) >= 86% and <90%

(B) >= 83% and <86%

(B-) >= 80% and <83%

(C+) >= 76% and <80%

(C) >= 73% and <76%

(C-) >= 70% and <73%

(D) >= 60% and <70%

(F) < 60%

Curve grading is applied:
adjust individual student grades based on the overall performance of the class, rather than using a fixed grading scale.

Mid-term Exam (15%)

- Exam will be based on the course material covered in class.
- Thursday, October 16 (tentative and subject to change)
- 6:00pm to 8:00pm

Final Exam (25%)

This course has a comprehensive final exam.

Exam will be based on the course material covered in class.

Thursday, December 11(The date is final.)

5:30-7:30 PM

In-class quizzes (10%)

- Students will be given in-class quizzes that will test their understanding and retention of presented materials.
- Students are required to submit their own work and cannot submit work for others.
- The lowest score of the quizzes (only one) would be dropped and not counted.
- *Students who submit in-class activity assignment including survey, but were not in class, as well as the students who submit on someone else's behalf are violating the academic integrity policy and will be reported immediately to the office of Student Conduct and Ethical Development.*

Assignments (10%)

- All assignments are needed to submit via Canvas by the deadline.
- If anything goes wrong, please contact ISA or the instructor. It is a student's responsibility to figure out if the instruction of assignments is not clear for him/her.
- You can submit as many times as you'd like until the deadline; we will grade only the last submission.
- All assignments might be re-evaluated at any time during the semester.

Late days for Assignments

- Each student has 6 late days to use.
- One late day extends the deadline by 24 hours.
- Each student can use up to 3 late days per assignment.

Optional Assignment

- There are some optional assignments.
- Score of optional assignment is not counted for your grade of Assignments.
- If it is submitted, the score of the assignment will be replaced with your assignment with the lowest score.

Regrade requests

- If you believe that the course staff made an objective error in grading in (quiz, Assignment, mid-term exam, final exam), you may request to regrade within 3 days after the grades are released.

Group project (20%)

- Groups of 3 - 4 students will be formed to work on a semester-long project.
- The project should be based on the course materials covered in class. Some points would de deducted for any content which not covered in class.
- Each group member is expected to participate in every phase of the project.
- The project will be graded based on the following components:

  a) progress check-up 5pts

  b) final presentation 10pts

  c) technical report 5 pts

Criteria for Group project

- The project/presentation/report would be reviewed according to the following criteria:
  - Community contributions
  - Any new or breakthrough work
  - Originality of concept/approach and level of innovativeness
  - Quality of research design/theoretical argument
  - Relevance to the course
  - Presentation and Writing style: its coherence and clarity of structure
  - Presentation and Writing style: Is easily understandable and represented clearly

Technical report for Group project

- Six-page (for double column) or ten-page (for single column) not counting references
- It should include:
  - Title
  - Authors
  - Abstract
  - Introduction: Description of purpose, motivation
  - Related works (optional)
  - Method(s)
  - Results: Supporting images/tables/figures
  - Conclusions

Group project schedule

tentative and subject to change

- Aug 28[2]: Group formation (on canvas)
- Sept 11[4]: Project proposal (within 5 mins)
- Oct 9 [8]: Project progress checkup (within 10 mins) – 5 pts
- Nov 13 [13] : Submit presentation slides/source codes/report (within 6 pages) – 5 pts
- Nov 13 [13] , 20 [14] : Project presentation (within 20 mins) – 10 pts

Lab(20%)

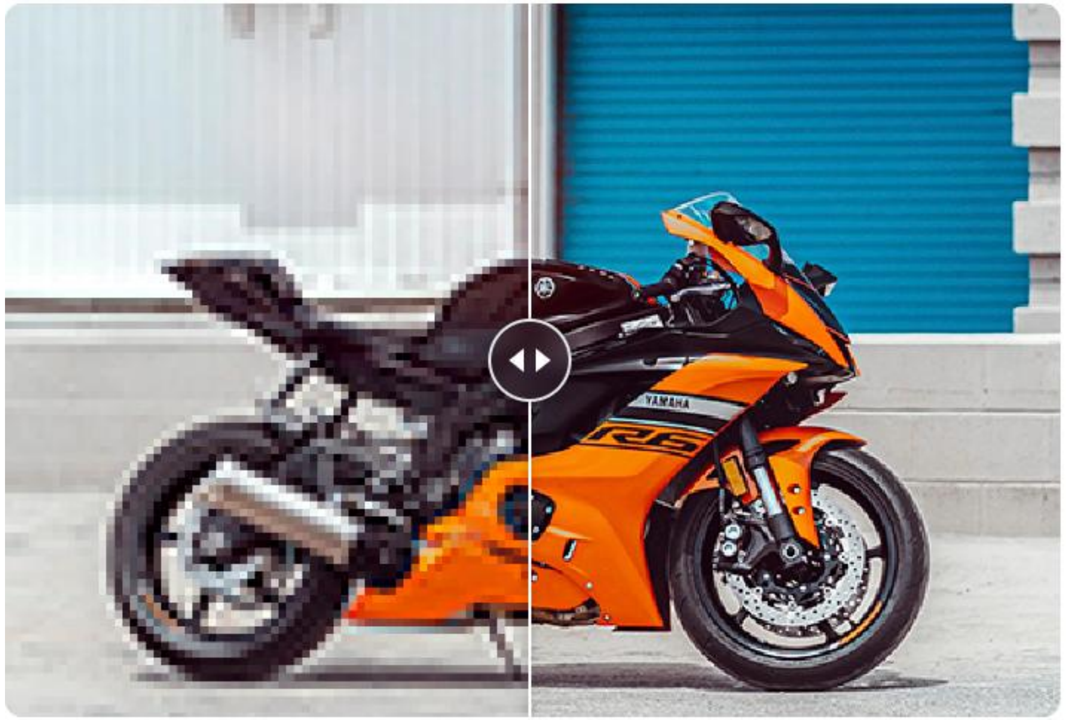
- Individual student will work on a semester-long project with the given subject.
- The project will be graded based on score metrics in benchmark result.
- $Score\ metric = \left(1 + 0.5 \times \log\left(\frac{S_{best}}{S}\right)\right) \times 20$
- You can submit as many times as you'd like until the deadline; we will grade only the last submission.
- The final submission deadline is November 20^{th} .
- Presentation will be on December 4^{th}.

Schedule

Wk	Date	Description
1	21-Aug	L1. Overview
2	28-Aug	L2. Generative AI, Fundamental Deep neural networks
		* Grouping for Group project
3	4-Sep	L3. Activation Functions, Initialize Weights, Forward Pass, Loss function
4	11-Sep	L4. Model Optimization Procedure, Gradient Descent, backward pass for binary classification
		* Group project proposal
5	18-Sep	L5. Backward pass for multiclass classification, Learning rate, Optimizer
6	25-Sep	L6. Operations in CNN
7	2-Oct	L7. CNN architectures for image classification, Data augmentation, Regularization for Overfitting
8	9-Oct	* Project progress checkup
9	16-Oct	* Mid-term Exam
10	23-Oct	L8. Object detection
11	30-Oct	L9. Semantic Segmentation, Autoencoder, VAE
12	6-Nov	L10. GAN, Clip, Diffusion models
13	13-Nov	* Project report/codes/slides submission
		* Project Presentation I
14	20-Nov	* Lab report/codes/slides submission
		* Project Presentation II
15	27-Nov	NO class (Thanksgiving break)
16	4-Dec	* Lab Presentation
17	11-Dec	* Final Exam

Lab topic: Super resolution

- Super Resolution uses machine learning to clarify, sharpen, and upscale the photo without losing its content and defining characteristics.
- Blurry images are unfortunately common and are a problem for professionals and hobbyists alike.
- Super resolution uses machine learning techniques to upscale images in a fraction of a second.



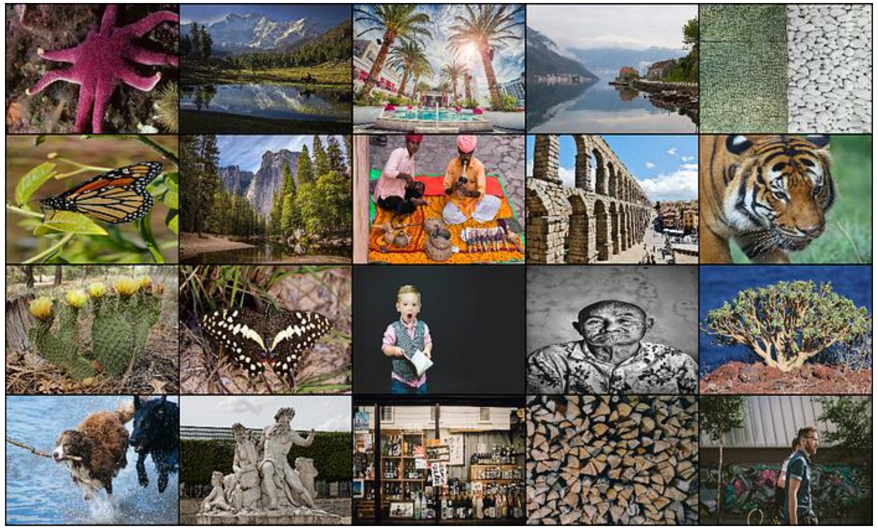
Lab: Super resolution

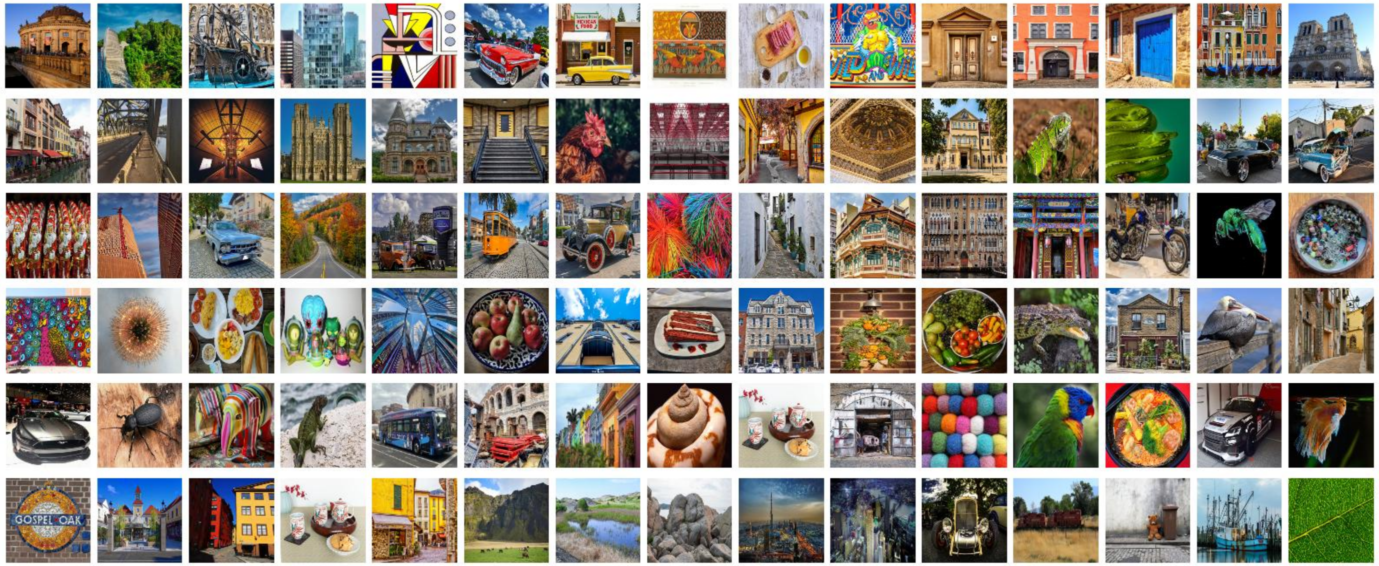
- Core Challenge:
  - Low-resolution (LR) images lose detail, gain noise, and artifacts from downscaling or degradation.
  - Traditional upscaling (e.g., Bicubic Interpolation) fails to recover details, adds blur
- Impact:
  - Hurts applications like medical imaging (e.g., missing MRI lesions) and surveillance (e.g., unclear license plates).
  - Limits low-cost imaging devices in professional use.
- Objective:
  - Develop a computationally efficient super-resolution model that effectively restores fine details and mitigates noise in degraded low-resolution images.

Lab: Super resolution

- Data set:
- DIV2K
  - https://data.vision.ee.ethz.ch/cvl/DIV2K/
  - 800 (train), 100 (valid)
- Flickr2K: 2,650
- LSDIR
  - https://ofsoundof.github.io/lsdir-data/
  - 84991 (train), 250 (valid)

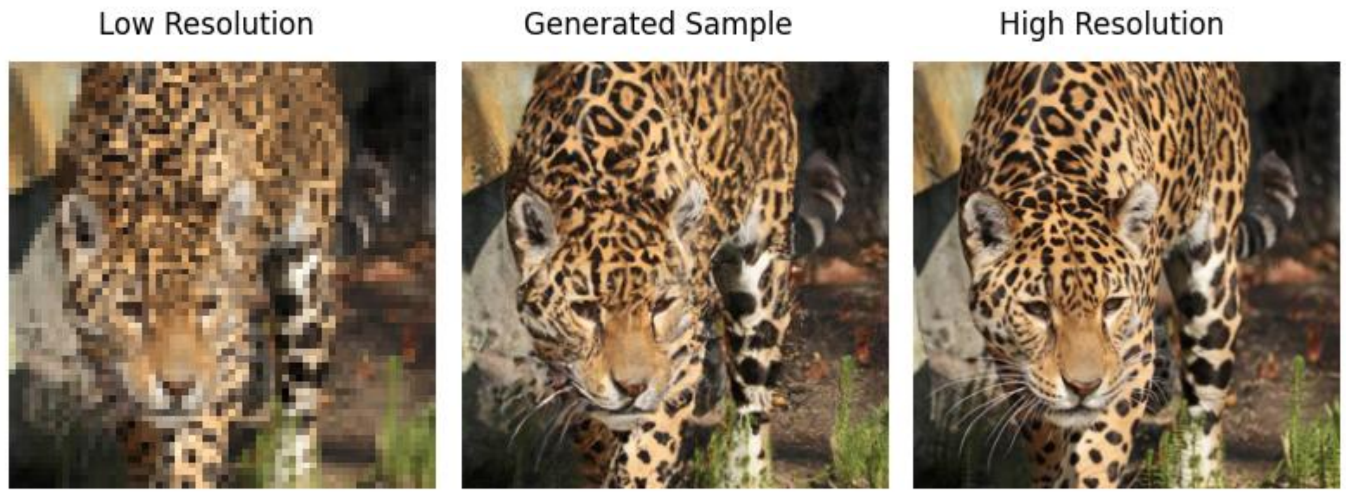
Custom valid set will be provided.
Do not use original valid sets for training.
It will be considered cheating.





Lab: Metrics

PSNR



Runtime

Lab: Evaluation

- Modified from NTIRE 2024 Efficient Super-Resolution Challenge
- $Score\ metric = \left(1 + 0.5 \times \log\left(\frac{S_{best}}{S}\right)\right) \times 20$
- $Score_{final} = w_1 \times Runtime + w_2 \times FLOPs + w_3 \times Parms$
  - $w_1: 0.7, w_2: 0.15, w_3: 0.15$
- PSNR should be >=26.90dB. Otherwise, score will be 10.
- PSNR, Runtime is average of valid and test set

runtime[ms]	FLOPs[G]	#Parms[M]	weighted sum	score
5.59	0.151	9.83	5.41	20
8.37	0.215	13.05	7.85	18.4
9.38	0.218	11.95	8.39	18.1
9.63	0.212	13.86	8.85	17.9
10.02	0.202	12.17	8.87	17.9
10.62	0.165	9.75	8.92	17.8
10.49	0.263	16.2	9.81	17.4
11.49	0.163	9.78	9.53	17.5
11.26	0.208	11.99	9.71	17.5
11.88	0.138	8.16	9.56	17.5
11.25	0.266	16.34	10.37	17.2
11.9	0.245	14.97	10.61	17.1
12.19	0.318	19.7	11.54	16.7
12.53	0.298	18.67	11.62	16.7
13.57	0.238	14.47	11.71	16.6
12.49	0.38	24.81	12.52	16.4
16.03	0.084	4.97	11.98	16.5
18.48	0.093	5.89	13.83	15.9
16.77	0.433	27.05	15.86	15.3
17.35	0.657	41.21	18.43	14.7
27.87	0.09	5.81	20.39	14.2
37.69	0.424	20.17	29.47	12.6

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Lab: Metrics

- Runtime is average of measured values for valid and test set on MLA100 (NPU PCA card)
    - Exclude top 5 and bottom 5 values from measured values
- PSNR is average of measured values for valid and test set on MLA100

Sponsor

- We would like to thank mobilint[1] for sponsoring for this class

MLA100 Low Profile

NPU PCIe Card

ARIES Specifications

Performance	Host Interface	Memory Capacity
80 TOPS	PCIe Gen 4.0 8-lane	16 GB (Optional 32 GB) LPDDR4, 4X
Memory Bandwidth	Power (TDP)	AI Performance
66.7 GB/s	25 W	MobileNetV2: 11,551 FPS
AI Performance	AI Performance	AI Performance
ResNet-50: 3,082 FPS	YOLOv9-S: 627 FPS	YOLOv9-C: 254 FPS

[1]https://www.mobilint.com/